

The Data Modeling Process

30-Minute Hands-On Exercise | Week 9 - Section 2A

Learning Objectives: Define a data model and explain why modeling choices matter | List and sequence the data modeling stages in Power BI | Distinguish between OLTP and OLAP systems | Identify what makes a source an ideal starting point for a Power BI model

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2A.

1.	A data model describes the <u>structure</u> of your data and how each part relates to the others.
2.	The first stage of the modeling process is to <u>discover</u> the source data by learning its existing shape and which values will be useful.
3.	In relational databases, the relationship between two tables is frequently recorded using a <u>foreign key</u> .
4.	A database optimized for reporting, structured with star or snowflake schema, and commonly called a Data Warehouse is known as an <u>OLAP</u> system.
5.	The ideal Power BI model organizes data into a <u>Star</u> schema.
6.	Adding calculations, hierarchies, and date attributes to the model is called the <u>augment</u> stage.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term	Description
A. Data Model >> 5	1. The original shape of data before it has been modeled or transformed
B. OLTP System >> 6	2. A constraint in relational databases that points to a related row in another table
C. OLAP System >> 3	3. A database structured for reporting; less normalized, uses star/snowflake schema
D. Source Model >> 1	4. The ideal target shape for a Power BI semantic model

E. Destination Model >> 4

F. Foreign Key >> 2

5. The shape that describes relationships within a dataset and how each part relates

6. Processes many simple, fast transactions; highly normalized; multi-user; emphasizes data integrity

Part 3: Modeling Stages in Order (~8 minutes)

The six data modeling stages are listed below in order. For each stage, write at least two key actions or questions that belong to that stage in the right-hand column.

Order	Modeling Stage	Key Action(s) in this Stage
1	1. Discover	Explore the data Identify which data is useful for reporting
2	2. Bring In (Ingest)	Load data into Power BI Connect to other sources like Excel or SQL
3	3. Profile	Inspect data Identify missing values or look for duplicates
4	4. Transform	Reshape tables Rename columns, change the structure to help with analysis
5	5. Clean	Correct typos Handle missing values and fix any problems in the data
6	6. Augment	Add calculated columns Add intelligence to the model

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A junior analyst receives a sales CSV file and immediately starts building visuals. What critical step(s) of the modeling process are they skipping, and why does that matter?

The analyst is skipping the Profile, Transform, and Clean stages. If the data contains errors, missing values, incorrect data types, or poor relationships, the visuals may produce misleading results. Understanding and preparing the data first helps ensure accurate analysis.

Question 2

Your source data uses the data type 'Float' for a price column. You want report users to see values displayed as currency (e.g., \$1,200.00). At which modeling stage would you address this, and why is Power BI better suited for it than a traditional relational database?

This would be handled during the Augment stage. Power BI allows analysts to apply business-friendly formatting such as currency, percentages, and date formats without changing the underlying data. A traditional relational database is primarily focused on storing data, while Power BI is designed for reporting and presentation.

Question 3

What is the difference between a 'simple' calculation and an 'advanced' calculation in the Augment stage? Give a concrete example of each.

A simple calculation performs basic arithmetic on existing fields. For example, creating a calculated column called Profit = Revenue - Cost. An advanced calculation uses DAX functions and context-aware logic. For example, a Year-to-Date Sales measure that changes automatically based on report filters and dates.

Question 4

Your company's data lives in a highly normalized OLTP database with 40 tables. A colleague suggests connecting Power BI directly to it without any modeling changes. What two problems could arise, and what would be a better approach?

One problem is that the large number of tables can make relationships difficult to manage. Another problem is slower report performance because Power BI may need to join many tables during analysis. A better approach is to transform the data into a reporting-friendly star schema or data warehouse structure before building the Power BI model.

Bonus / Reflection (if time allows)

Section 2A states: 'Understanding your source data is essential to the modeling process -- and understanding an ideal destination model is just as essential.' In 3-5 sentences, explain what this means in practice. Why is it not enough to understand just one side?